CS 2840 Fall 2025 Scribes: Jade Nair and Eliza Knapp
10/09

Sliced Optimal Transport with Kimia Nadjahi

1 Reminders on Optimal Transport

1.1 The First OT Problem (Monge, 1781)

The first optimal transport problem was posed by Monge in 1781.

> How do you move sand into a pit at minimal cost?

This problem has been applied in several machine learning fields, one of which is Optimal Transport (OT). OT is a versatile tool for comparing probability distributions. It aims to create an assignment between two distributions—either discrete or continuous—by minimizing a notion of distance between their mass.

For these notes, this distance will be the **Wasserstein distance**, defined below.

Wasserstein distance:

$$W_2(\mu, \nu) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{X \times Y} \|x - y\|^2 \, d\pi(x, y)$$

Here, μ and ν are two probability measures (distributions) on spaces X and Y, and $\Pi(\mu, \nu)$ denotes the set of all *couplings* π whose marginals are μ and ν.

If the measures admit densities with respect to the Lebesgue measure, we can write

$$d\mu(x) = \rho_\mu(x)\, dx, \qquad d\nu(y) = \rho_\nu(y)\, dy,$$

where ρ_μ and ρ_ν are the corresponding probability density functions.

In many practical settings, one of the distributions is discrete. For example, we may represent ν as a finite weighted sum of Dirac masses:

$$\nu = \sum_{j=1}^{m} b_j \, \delta_{y_j},$$

where each y_j is a point in the target space and b_j is the associated weight or probability mass.

Interpretation:

- $\rho_\mu(x)$ — continuous source density (e.g., a smooth distribution of sand).
- $\rho_\nu(y)$ — continuous or discrete target density (e.g., pit locations).
- $\pi(x, y)$ — transport plan specifying how mass at x is moved to y.

In the case where the Monge problem (finding a deterministic transport map) is impossible to solve, we instead use the **Kantorovich relaxation**, which allows probabilistic couplings between μ and ν. 3

1.2 Optimal Transport for Machine Learning

Generative modeling provides a natural setting for applying Optimal Transport (OT). We can summarize the goal of generative modeling as follows:

Given a set of observed data points (our target distribution), we want to learn a transformation that maps a simple *base distribution* (e.g., a standard Gaussian) into a complex distribution that closely approximates the data.

Formally, we define a transformation $T_{\mathbf{w}}$ parameterized by neural network weights $\mathbf{w}$. Applying $T_{\mathbf{w}}$ to samples from a simple source distribution μ_0 induces a pushforward distribution:

$$\mu_{\mathbf{w}} = (T_{\mathbf{w}})_{\#}\mu_0.$$

The goal of training is to find the parameters $\mathbf{w}$ that make $\mu_{\mathbf{w}}$ as close as possible to the empirical data distribution $\hat{\mu}_n$:

$$\min_{\mathbf{w}} D\big(\mu_{\mathbf{w}}, \hat{\mu}_n\big),$$

where D is a suitable measure of discrepancy between distributions (e.g., a divergence or distance).

Wasserstein GAN (WGAN). The first highly successful generative architecture to use Optimal Transport ideas was the *Wasserstein GAN (WGAN)*. Rather than relying on the Jensen–Shannon divergence of standard GANs, it reformulates the objective function in terms of the **dual form** of the Wasserstein-1 distance:

$$W_1(\mu_{\mathbf{w}}, \hat{\mu}_n) = \sup_{\|f\|_L \le 1} \mathbb{E}_{x\sim\mu_{\mathbf{w}}}[f(x)] - \mathbb{E}_{y\sim\hat{\mu}_n}[f(y)].$$

This formulation improved training stability and interpretability, showing that OT can provide meaningful gradients even when the support of the two distributions does not overlap. Following WGANs, a number of related models were proposed, such as:

- **Wasserstein Auto-Encoders (WAE):** Auto-encoders trained by minimizing a regularized Wasserstein distance between the encoded and target distributions.

Let us consider a simple one-dimensional example to understand the geometry of OT.

Let

$$\mu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}, \qquad \nu = \frac{1}{n}\sum_{i=1}^{n} \delta_{y_i}.$$

For one-dimensional distributions, the optimal transport simply pairs sorted samples:

$$W_2^2(\mu, \nu) = \frac{1}{n}\sum_{i=1}^{n} |x_{(i)} - y_{(i)}|^2,$$

where $x_{(1)} \le \cdots \le x_{(n)}$ and $y_{(1)} \le \cdots \le y_{(n)}$.

In one dimension, therefore, computing the Wasserstein distance reduces to sorting both sets of points along the real line and pairing them in order—an elegant and computationally efficient property.

1.3 Optimal Transport in Multiple Dimensions

We now extend Optimal Transport to the case where both distributions are defined on $\mathbb{R}^d$.

Let

$$\mu = \sum_{i=1}^{n} a_i \, \delta_{x_i}, \qquad \nu = \sum_{j=1}^{m} b_j \, \delta_{y_j},$$

be two **discrete probability measures** on $\mathbb{R}^d$, where

$$a_i \geq 0, \ \sum_{i=1}^{n} a_i = 1, \qquad b_j \geq 0, \ \sum_{j=1}^{m} b_j = 1.$$

The goal is to find a *transport plan* $\pi = (\pi_{ij})_{i,j}$ indicating how much mass is moved from source point x_i to target point y_j, so that the total cost is minimized.

$$W_2^2(\mu, \nu) = \min_{\pi \in \Pi(a,b)} \sum_{i=1}^{n} \sum_{j=1}^{m} c_{ij} \, \pi_{ij},$$

where

$$c_{ij} = \|x_i - y_j\|^2$$

defines the **cost matrix** $C \in \mathbb{R}^{n \times m}$, and $\Pi(a, b)$ is the set of all couplings satisfying the marginal constraints:

$$\sum_{j=1}^{m} \pi_{ij} = a_i, \qquad \sum_{i=1}^{n} \pi_{ij} = b_j, \qquad \pi_{ij} \geq 0.$$

Discrete OT as a Linear Program. The finite-dimensional OT problem is a *linear program (LP)*:

- Variables: $n \times m$ transport coefficients π_{ij}
- Constraints: $n + m$ marginal constraints
- Objective: minimize the linear cost $\langle C, \pi \rangle$

Solving this LP yields an optimal coupling between the two discrete distributions. However, since the number of variables grows quadratically with the number of samples, the computational cost can become prohibitive for large n. This motivates the development of faster approximations such as **entropic regularization** and the **Sinkhorn algorithm**, which we will discuss next.

1.4 Standard OT: Too Costly in Machine Learning

While Optimal Transport provides a powerful framework for comparing distributions, it suffers from significant **computational limitations** in practice.

Computing the discrete Wasserstein distance $W_2(\mu_n, \nu_n)$ exactly requires solving a linear program with n^2 variables and $2n$ constraints. The best exact solvers scale as $O(n^3)$, which quickly becomes intractable for modern machine learning applications.

Example: OT-based Neural Network Training. Suppose we train a neural network that uses the Wasserstein distance as part of its loss. Even for a moderate mini-batch size of $n = 256$, one evaluation of $W_2(\mu_n, \nu_n)$ already involves roughly 10^7 operations. Since this computation must be repeated at every training step—often over thousands of iterations— the total cost becomes prohibitive.

As a result, most practical applications rely on **approximate** or **regularized** versions of OT that trade off a small amount of bias for massive computational savings.

Approximate / Regularized OT Methods

- **Entropy-Regularized OT (Sinkhorn Algorithm)** Introduced by Cuturi (2013). Adds an entropy term to the OT objective, making the problem strictly convex and enabling fast matrix-scaling iterations. [Cut13]
- **Sliced OT (Rabin et al., 2011) Projects high-dimensional distributions onto one-dimensional subspaces, computes 1-D OT distances (which are fast to evaluate), and averages over random directions. [JPDM11]**

Both methods significantly reduce computational cost, allowing OT to be incorporated into deep learning pipelines and large-scale generative models.

1.5 The Curse of Dimensionality

In practice, we never have direct access to the full underlying distributions μ and ν. Instead, we observe only finite samples:

$$(x_i)_{i=1}^n \sim_{\text{i.i.d.}} \mu, \qquad (y_i)_{i=1}^n \sim_{\text{i.i.d.}} \nu.$$

Thus, we can only compute Wasserstein distances between their **empirical measures**:

$$\mu_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}, \qquad \nu_n = \frac{1}{n}\sum_{i=1}^n \delta_{y_i}.$$

We compare the empirical distance $W_p(\mu_n, \nu_n)$ to the true population distance $W_p(\mu, \nu)$. As $n \to \infty$, the empirical Wasserstein distance converges to the true one, but the rate of convergence depends strongly on the dimension d.

Convergence Rate. If μ and ν are supported on a bounded domain in $\mathbb{R}^d$ with $d > 2$, then the expected convergence rate is:

$$\mathbb{E}\left[\, |W_p(\mu_n, \nu_n) - W_p(\mu, \nu)| \,\right] = O\left(n^{-\frac{1}{d}}\right),$$

where the expectation is taken over all samples $(x_i, y_i)_{i=1}^n$.

Consequences:

- As the dimensionality d increases, convergence becomes dramatically slower.
- Achieving accurate estimates may require prohibitively large sample sizes.
- Increasing n also increases computational cost, since exact OT scales superlinearly in n.

This phenomenon is known as the **curse of dimensionality**. In high-dimensional ML settings, one must therefore balance *sample complexity* and *computational complexity*. These challenges motivate approximate and low-dimensional alternatives, such as **sliced OT** and **projection-based methods**.

2 Sliced Optimal Transport

2.1 Slicing: From High to One Dimension

The key idea behind **Sliced Optimal Transport (SOT)** is to reduce a high-dimensional OT problem to a collection of one-dimensional OT problems, which can be solved efficiently.

Let

$$\theta = \begin{bmatrix} \theta^{(1)} \\ \theta^{(2)} \\ \vdots \\ \theta^{(d)} \end{bmatrix} \in \mathbb{R}^d, \qquad x = \begin{bmatrix} x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(d)} \end{bmatrix} \in \mathbb{R}^d.$$

We define the **projection of a point** x onto the direction θ as

$$\langle \theta, x \rangle = \sum_{j=1}^{d} \theta^{(j)} x^{(j)} \in \mathbb{R}.$$

That is, each point in $\mathbb{R}^d$ is projected onto a scalar coordinate along the direction θ.

One-Dimensional Projections. For a given direction $\theta \in \mathbb{R}^d$, we define the one-dimensional projections of the measures μ and ν as

$$\mu^\theta = (P_\theta)_\# \mu, \qquad \nu^\theta = (P_\theta)_\# \nu,$$

where $P_\theta(x) = \langle \theta, x \rangle$ is the projection map and $(\cdot)_\#$ denotes the pushforward measure.

Since one-dimensional Wasserstein distances can be computed in closed form via sorting, the idea is to average such distances across many random directions.

Definition (Sliced Wasserstein Distance). For $p \geq 1$ and two d-dimensional probability measures μ, ν, the *Sliced Wasserstein distance* is defined as

$$SW_p(\mu, \nu) = \left(\mathbb{E}_{\theta \sim U(S^{d-1})} \left[W_p^p(\mu^\theta, \nu^\theta) \right] \right)^{1/p},$$

where $U(S^{d-1})$ denotes the uniform distribution over the unit $(d-1)$-sphere

$$S^{d-1} = \{\theta \in \mathbb{R}^d \, : \, \|\theta\|_2 = 1\}.$$

2.2 Connection to the Radon Transform

The **Sliced Wasserstein distance** can be expressed naturally using the **Radon Transform** of measures.

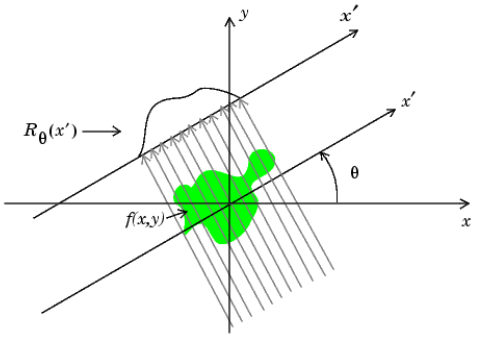


Figure 1: Radon Transform illustration: projecting a high-dimensional measure onto 1D lines along direction θ.

For each direction $\theta \in \mathbb{R}^d$, the Radon transform projects a measure onto a line orthogonal to θ, yielding a one-dimensional object. This process is central to many imaging applications, such as **tomography** and **CT scans**, where an image is reconstructed from its projections.

Sliced Wasserstein via the Radon Transform.

$$SW_p(\mu, \nu) = \left(\mathbb{E}_{\theta \sim U(S^{d-1})}\left[W_p^p(\mu^\theta, \nu^\theta)\right]\right)^{1/p},$$

where each one-dimensional marginal (μ^θ, ν^θ) is obtained by projecting the high-dimensional measures (μ, ν) along θ.

Intuitively, to compute SW_p, we:

1. Sample directions $\theta_1, \ldots, \theta_k \sim U(S^{d-1})$.
2. Project μ and ν onto each direction θ_j to obtain μ^{θ_j} and ν^{θ_j}.
3. Compute the one-dimensional Wasserstein distance $W_p(\mu^{\theta_j}, \nu^{\theta_j})$.
4. Repeat for k directions and average.

This yields the practical Monte Carlo approximation:

$$SW_p(\mu, \nu) \approx \left(\frac{1}{k}\sum_{j=1}^{k} W_p^p(\mu^{\theta_j}, \nu^{\theta_j})\right)^{1/p}.$$

If

$$\mu = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i}, \qquad \nu = \frac{1}{n}\sum_{i=1}^{n} \delta_{y_i},$$

then for each direction θ_j, the one-dimensional Wasserstein distance simplifies to

$$W_p^p(\mu^{\theta_j}, \nu^{\theta_j}) = \frac{1}{n}\sum_{i=1}^{n} \left|\langle \theta_j, x_{(i)} \rangle - \langle \theta_j, y_{(i)} \rangle\right|^p,$$

where the samples $\{x_{(i)}\}$ and $\{y_{(i)}\}$ are sorted along the projection direction θ_j.

Computational Complexity.

$$O(kdn + kn \log n)$$

- kdn: cost of projecting n points in d dimensions onto k directions
- $kn \log n$: cost of sorting projected points

In contrast, the standard OT problem scales as $O(n^3)$. Thus, Sliced OT is far more efficient when $n \gg d$ and k is moderate.

Questions.

Q: Are there more strategic ways to sample directions? For instance, could we use principal components instead of random θ's?

A: Yes — we can tweak this definition to include *non-uniform distributions* over directions. This will be discussed later.

Q: How do we guarantee that Sliced OT scales similarly to the true Wasserstein distance?

A: This question is also addressed later on in the lecture (it involves theoretical bounds on the approximation quality of SW_p to W_p).

2.3 From Color Transfer to Sliced OT

The origins of Sliced Optimal Transport trace back to the field of **image processing**, where related ideas were first developed for color transfer.

2006 — Iterative Distribution Transfer (IDT): Introduced as an algorithm for *color transfer* between images by iteratively matching one-dimensional projections of color distributions.

- Matches color histograms along several 1D projections.
- Conceptually similar to Sliced OT, but projections are restricted to an **orthonormal basis**.
- Developed by Pitié, Kokaram, and Dahyot (2007) [PKD07].

2009-2012 — Sliced Wasserstein Distance: Bernot, Caselles, and Morel generalized the IDT idea to a full probabilistic framework, defining the *Sliced Wasserstein distance* between arbitrary distributions. Originally introduced in a different context—computing Wasserstein *barycenters*—the formulation emphasized computational benefits and interpretability. [BCM09]

2013 — Theoretical Foundations: In his Ph.D. thesis, **Nicolas Bonnotte** provided a theoretical analysis of Sliced OT and established its connections to **gradient flows** in the Wasserstein space. This formalized Sliced OT as a mathematically grounded and computationally tractable approximation to the full OT distance. [Bon13]

Together, these developments transformed an image-processing heuristic into a general-purpose tool for large-scale machine learning, computer vision, and generative modeling.

2.4 ...To Generative Modeling

The ideas of Sliced Optimal Transport resurfaced a decade later within the field of **generative modeling**, where Wasserstein-based losses became central to training stable and geometry-aware neural networks.

2017 — Wasserstein GANs and Auto-Encoders. The introduction of *Wasserstein GANs (WGANs)* and *Wasserstein Auto-Encoders (WAEs)* demonstrated that OT-based distances could stabilize training and produce more meaningful gradients for generative models. [ACB17] [TBGS19]

2018 — Sliced-Wasserstein GANs and Auto-Encoders. Deshpande *et al.* (2018) proposed *Sliced-Wasserstein GANs* and related auto-encoder architectures, which replace the full Wasserstein distance with its sliced approximation.

- Retain much of the geometric fidelity of OT.
- Dramatically reduce computational cost.
- Show improved robustness compared to WGANs, especially in high-dimensional settings.

[DZS18]

Since 2018, this line of work has **renewed broad interest in Sliced Optimal Transport**, leading to numerous extensions and applications across deep generative modeling, domain adaptation, and representation learning.

2.5 Mathematical Properties of Sliced Wasserstein Distance

The following few sections reference the foundational papers mentioned above and are further extended in the citations in section 2.9.

The Sliced-Wasserstein distance satisfies all **metric axioms**:

For $\mu, \nu, \epsilon \in P(R^d)$:

Non-negativity: $SW_p(\mu, \nu) \geq 0$

Identity of indiscernibles: $SW_p(\mu, \nu) = 0 \Leftrightarrow \mu = \nu$

Symmetry: $SW_p(W_p(\mu, \nu) = SW_p(\nu, \mu)$

Triangle inequality: $SW_p(\mu, \nu) \leq SW_p(\mu, w) + SW_p(w, \nu)$

2.6 Comparison with the Wasserstein distance

There are several ways to compare and connect the sliced Wasserstein distance with the original Wasserstein distance. The following show that the sliced Wasserstein distance is closely connected to the original, often differing by constant factors, and therefore often displays similar properties.

Always smaller: The Sliced Wasserstein distance is always less than or equal to the Wasserstein distance multiplied by a factor c:

$$\forall \mu, \upsilon \in P(\mathbb{R}^d), SW_p^p(\mu, \upsilon) \leq c_{d,p} W_p^p(\mu, \upsilon)$$

with $c_{d,p} = \frac{1}{d} E_{\theta \sim U(S^{d-1})}[||\theta||^p] \leq 1$ if $p \geq 2$, $c_{dp} \leq \frac{1}{d}$.

Equivalent on bounded supports: If $\mu, \upsilon \in B(0, R)$ where B is a ball of radius R, then

$$SW_p^p(\mu, \upsilon) \leq c_{d,p} W_p^p(\mu, \upsilon) \leq C_{d,p} R^{\frac{p-1}{d+1}} SW_p(\mu, \upsilon)^{\frac{1}{d+1}}$$

Exact formula for Gaussians: If $\mu = N(m_1, \sigma_1^2 I_d)$, then

$$SW_2(\mu, \nu) = \frac{1}{\sqrt{d}} W_2(\mu, \nu)$$

2.7 Slicing beats the curse of dimensionality

Next, consider the curse of dimensionality and complexity issues we discussed above. We can now show that the sliced distance SW_2 avoids some of these pitfalls by moving the problem into a lower-dimensional space.

To demonstrate, compare SW_2 to W_2 if you take each between two sets of n i.i.d. samples from $N(0, 1)$. When computing distance between two sets of samples, see that the one-dimensionality of the sliced OT helps make the convergence rate more manageable. This allows for us to avoid the tradeoff between sample and computational complexity as dimensionality increases, avoiding the large computational costs and/or sample sizes required for accuracy under the original Wasserstein distance.

Convergence rate of W_p (as in section 1.5):

$$\mathbb{E}\big[\,|W_p(\mu_n, \nu_n) - W_p(\mu, \nu)|\,\big] = O\Big(n^{-\frac{1}{d}}\Big),$$

Convergence rate and sample complexity of SW_p:
For μ, ν supported on a bounded domain of $\mathbb{R}^d$,

$$E(SW_p(\mu_n, \nu_n)) - SW_p(\mu, \nu) = O(n^{-\frac{1}{2p}})$$

A major benefit of using sliced OT is that it allows us to use this **dimension free rate** for convergence, whereas with W_p, the complexity of $O(n^{-1/d})$ scales with the dimension d, making it less tractable as the dimension increases.

2.8 Still some practical challenges

Despite the solution sliced OT presents to the dimensionality problem, it still encounters some issues when used in practice. The previous rate shown above applies to $SW_p(\mu, \nu) = \mathbb{E}(\theta)$ distributed as $U(\mathbb{S}^d - 1)$ of $W_p^p(\mu_\theta, \nu_\theta)$ to the $\frac{1}{p}$, which is intractable in practice.

Instead, the practical approximation uses k random projections:

$$\hat{SW}_p^p(\mu, \nu) = \frac{1}{k} W_p^p(\mu_j^\theta, \nu_j^\theta)$$

We can then determine the existence of a Monte Carlo approximation error, to understand the accuracy tradeoff of this approach.

Monte Carlo Approximation Error:

$$E[|\hat{SW}_p^p(\mu, \nu) - SW_p^p(\mu, \nu))|] \leq \frac{1}{k} var_\theta W_p^p(\mu^\theta, \nu^\theta)$$

This approximation presents a central tradeoff: more projections mean **better accuracy**, but **higher computational cost**.

Question: If we want to use sliced Wasserstein to approximate the true Wasserstein distance, in the Gaussian setting there's a constant factor of difference. If we account for that factor, can the sliced Wasserstein distance serve as a good approximation of the true Wasserstein distance in general? Are there any guarantees that if you have the wrong distances, the approximation wouldn't be very bad?

Answer: No. The sliced Wasserstein distance shouldn't be considered as another approximation to Wasserstein; rather it's an alternative distance that's inspired by the optimal transport framework. In special cases like Gaussian distributions, you have a matter of scaling between the two, but in general this isn't true. **If your goal is to actually compute the Wasserstein distance, slicing is not enough.**

Question: Can we find a better approximation through max slicing by using the supremum of the sliced data instead of the expectation?
Answer: Yes, the max slice gets us closer to the Wasserstein distance. This approach is mainly used when you're in a high dimensional setting and it's hard to find the useful directions that distinguish the distributions. Using the supremum as an approximation, you get the maximum that accomplishes your goal, because you are trying to find the direction that differentiates the most from the real data. This way, you can correct for it and get closer to the distribution.

2.9 Beyond the classical Sliced-Wasserstein distance

$$SW_p(\mu, v) = [E_{\theta \sim U(\mathbb{S}^{d-1})} W_p^p(\mu^\theta, v^\theta)]^{\frac{1}{p}}$$

The following additional papers explore extensions of this work in several different categories.

- Beyond Uniform Sampling
  - Max Sliced Wasserstein(Deshpande et al., 2019 [DHS+25], Paty and Cuturi., 2019 [PC19])
  - Distributional and energy-based Sliced Wasserstein (Nguyen et al., 2021 [NHPB20]; Nguyen and Ho., 2023 [NH23])
- Beyond Linear Projections
  - Generalized Sliced Wasserstein (Kolouri et al., 2019 [KNS+19])
  - On Manifolds:
    - Sphere (Bonet et al. [BBC+23], 2023; Tran et al., 2024[TBK+24]; Quellmalz et al., 2024 [QBS24])
    - Cartan-Hadamard manifolds (Bonet et al., 2025) [BDC24]
- Beyond Monte Carlo Estimation
  - Orthonormal sampling (Wu et al., 2019 - [WHA+19]; Rowland et al., 2019 [RHT+19])
  - Quasi-Monte Carlo (Nguyen et al., 2024 [NBH24]; Sisouk et al., 2025 [SDT25])
  - Variance reduction via control variates (Leluc et al., 2024 [LDP+24])
  - Concentration of measure approximation (Nadjahi et al., 2021) [NDJ+21]
- Broader Overviews (Nguyen, 2025 [Ngu25]; Sisouk et al., 2025 [SDT25])

2.10 Max-Sliced Wasserstein Distance

The max-sliced Wasserstein distance is a potential way to reduce the projection complexity introduced in earlier sections. It maximizes the sliced Wasserstein distance over $\theta \in \mathbb{S}^{d-1}$. $maxSW_p$ has specific topological properties and several methods by which it can be calculated.

Max-Sliced Wasserstein distance (Deshpande et al., 2019):

Let p ≥ 1 and μ, ν be two probability measures on $\mathbb{R}^d$:

$$maxSW_p(\mu, \nu) = max_{\theta \in \mathbb{S}^{d-1}} W_p(\mu^\theta, \nu^\theta)$$

Topological Properties of Max-SWp:

$$SW_p(\mu, \nu) \leq maxSW_p(\mu, \nu) \leq W_p(\mu, \nu)$$

$$\lim_{k \to \infty +} W_p(\mu^k, \mu) = 0 \Leftrightarrow \lim_{k \to \infty +} maxSW_p(\mu^k, \mu) = 0$$

for a sequence of measures $(\mu^k)k \in N$.

How to compute $maxSW_p(\mu, \nu)$:

- Projected gradient ascent (Kolouri et al., 2019; Nietert et al., 2022)
- Riemannian optimizations on $\mathbb{S}^{d-1}$ (Lin et al., 2020)

2.11 Generalized Sliced Wasserstein distances

In order to reduce computational complexity by using fewer projections, this method aims to **generalize the linear projections** $x \to \theta(x)$ used in regular Sliced Wasserstein.

With GSW, we can now use **general polynomial projection functions** $g^\theta : \mathbb{R}^d \to \mathbb{R}$ to define μ^θ, ν^θ.

Projection functions give different level sets from different n-dimensional surfaces.

Generalized Sliced-Wasserstein distances

Formally, this distance is represented as:

$$GSW_p(\mu, \nu) = [\mathbb{E}_{\theta \sim U(\mathbb{S}^{d-1})} W_p(g^\theta_\# \mu, g^\theta_\# \nu)]$$

Question: Are there sample complexity results for the generalized case?
Answer: I don't think there's work on that yet. It's a hard question, and if you rely on analytical predictions you could do something, but I don't think it's something people have done.

Question: Is it possible to show that with an arbitrarily rich class of projectors can recover exact Wasserstein distance?
Answer: There are no results yet in that area.

2.12 Transport Maps with Slicing - From Transport Distances to Transport Maps

So far, we have focused on sliced OT distances, yet in many applications, **the useful object is the transport map**. Returning back to the Monge problem:

Find a transport map $T : \mathbb{R}^d -> \mathbb{R}^d$ solving

$$T = argmin_{s_E \mu = \nu} E_{x \sim \mu}[||x - S(x)||^2]$$

Transport maps themselves have a variety of applications. Some of the domains in which they matter include:

- Domain adaptation/transfer learning (Courty et al., 2017)
- Computational biology (single cell genomics) (Schiebinger et al., 2019; Bunne et al., 2024)

- Multivariate quantile and ranks

Currently, sliced OT defines:

$$SW_p(\mu, v) = [\mathbb{E}_{\theta \sim U(\mathbb{S}^{d-1})} W_p^p(\mu^\theta, v^\theta)]^{\frac{1}{p}}$$

Resulting in the transport map:

$$T_{\theta \in S^{d-1}} \; s.t. \; (T_\theta) \# \mu_\theta = \nu_\theta$$

The following method can be applied to reconstruct a global $T : R^d \to R$ from $\theta \in \mathbb{S}^{d-1}$:

Slice Matching Maps: (Li and Moosmuller, 2024)

Let $P\theta_1 ... \theta_d \in \mathbb{R}^{dxd}$ be an orthonormal basis of $\mathbb{R}^d$:

$$\forall x \in R^d, T_p(x) = \sum T_i^\theta(\theta_i, x)(\theta_i)$$

where T_i^θ is a 1-D transport map between μ_i^θ and ν_i^θ

Recent alternatives also include:

- Pivot based Sliced Wasserstein plans
- Expected Sliced Wasserstein plans
- Theoretical analysis and numerical comparison

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